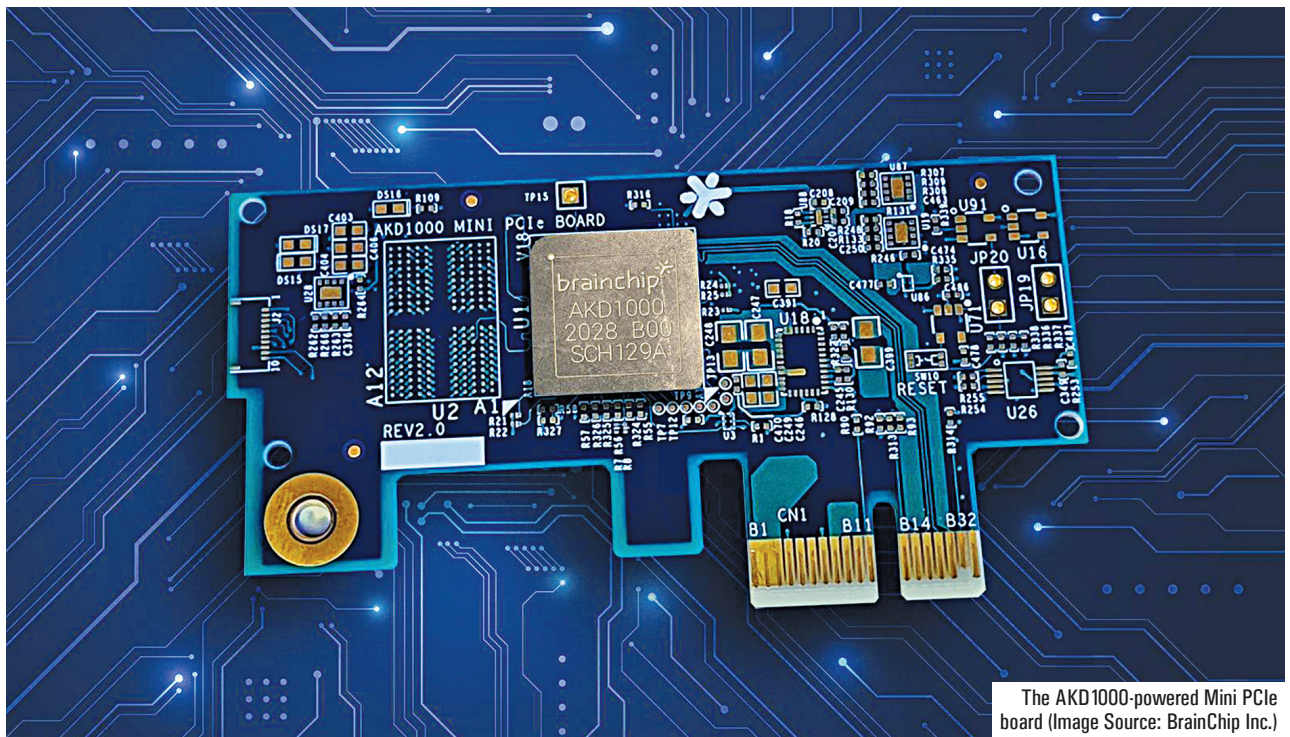


What's So Exciting About Neuromorphic Computing

The human brain is the most efficient and powerful computer that exists. Even after decades and decades of technological advancements, no computer has managed to beat the brain with respect to efficiency, power consumption, and many other factors. Will neuromorphic computers be able to do it?



The AKD1000-powered Mini PCIe board (Image Source: BrainChip Inc.)

The exact sequence of events that take place when we do a particular activity on our computer, or on any other device, completely depends on its inherent architecture. It depends on how the various components of the computer like the processor and memory are structured in the solid state.

Almost all modern computers we use today are based on the Von Neumann architecture, a design first introduced in the late 1940s. There, the processor is responsible for executing instructions and programs, while the

In his paper titled 'Neuromorphic Electronic Systems,' American scientist and engineer Carver Mead writes, "Biological information-processing systems operate on completely different principles from those with which engineers are familiar. For many problems, particularly those in which the input data are ill-conditioned, and the computation can be specified in a relative manner, biological solutions are many orders of magnitude more effective than those we have been able to implement using digital methods."

memory stores those instructions and programs. When you think of your body as an embedded device, your brain is the processor as well as the memory. The architecture of our brain is such that there is no distinction between the two.

Since we know for a fact that the human brain is superior to every single computer that exists, doesn't it make sense to modify computer architecture in a way that it functions more like our brain? This was what many scientists realised in the 1980s, starting with